

STATEMENT OF NEED (SON)

1.0 TITLE

Acquisition of Software for Whole-Genome Sequencing

2.0 BACKGROUND

Recent advances in next generation sequencing technologies have dramatically increased the rate of data output while significantly reducing costs. The Integrative Tumor Epidemiology Branch (ITEB), Division of Cancer Epidemiology and Genetics (DCEG), National Cancer Institute (NCI) is currently analyzing a large number of tumor and normal tissue samples of lung cancer and kidney cancer and plan to analyze additional cancer types in the future using whole-genome sequencing (WGS). However, highly accurate analysis of sequencing data is computationally intensive and creates a bottleneck in the overall sequencing workflow.

To address these challenges, ITEB is looking to procure a software that will significantly decrease the analysis time and the computational resource requirements for variant detection without compromising accuracy. The software will result in a 20-to-50-fold increase in processing speed on the same hardware with results that are identical to the currently used GATK pipeline. Due to the large sample size of tumors ITEB is analyzing, a software that can provide variant calling at high speed is essential for its ongoing and planned projects. Since this is a large-scale, complex, and ongoing analysis, ITEB anticipates needing the software for the next five years to complete the major analyses.

3.0 TYPE OF ORDER

This is a Severable Firm Fixed-Price Purchase Order.

4.0 SCOPE

The contractor shall be able to provide software necessary to provide variant calling at high speed.

5.0 SPECIAL ORDER REQUIREMENTS

5.1 PRODUCT FEATURES/SALIENT CHARACTERISTICS

The following product features/characteristics are required for this requirement:

- 4.1.1. 20-to-50-fold increase in processing speed for key tools such as BWA-mem, STAR, minimap2, Picard, and GATK while maintaining mathematical consistency.
- 4.1.2. Exceptional accuracy and reproducibility for next-generation sequencing data variant calling with no run-to-run variation.

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- 4.1.3. Process a 30x WGS sample data in under 30 minutes for less than \$2 compute cost.
- 4.1.4. Support joint calling of large-number of WGS directly from VCFs without intermediate merging to facilitate large-cohort studies.
- 4.1.5. Support multiple complex use cases including short and long reads, germline and somatic variant calling, structural variants, copy number variants.

5.2 DELIVERY / INSTALLATION

The software shall be delivered within (5) calendar days of purchase order award. The contractor that receives the award will be provided with the complete mailing address and point of contact. Upon delivery of the award, the contractor must notify the COR (TBD) to arrange delivery.

Point of Contact (TBD)

6.0 PERIOD OF PERFORMANCE

7.0 INSPECTION AND ACCEPTANCE

Pursuant to FAR clause 52.212-4, all material/supplies as described in the Statement of Need (SON) to be delivered under this purchase order is subject to final inspection and acceptance by an authorized representative of the Government.

8.0 PAYMENT

Payment shall be made upon receipt of software and approval of the COR. Payment authorization requires submission and approval of invoices to the Contracting Officer's Representative (COR) and National Institutes of Health, Office of Financial Management, in accordance with the payment provisions listed below:

The following clause is applicable to all Purchase Orders, Task or Delivery Orders, and Blanket Purchase Agreement (BPA) Calls: PROMPT PAYMENT (JAN 2017) FAR 52.232-25. Highlights of this clause and NIH implementation requirements follow:

I. Invoice Requirements

- A. An invoice is the Contractor's bill or written request for payment under the contract for supplies delivered or services performed. A proper invoice is an "Original" which must include the items listed in subdivisions 1 through 12 below, in addition to the requirements of FAR 32.9. If the invoice does not comply with these requirements, the Contractor will be notified of the defect within 7 days after the date the designated billing office received the invoice (3 days for meat, meat food products, or fish, and 5 days for perishable agricultural commodities, dairy products, edible fats, or oils) with a statement of the reasons why it is not a proper invoice. (See exceptions under II., below.) Untimely notification will be considered in the computation of any interest

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penalty owed the Contractor.

1. Vendor/Contractor: Name, Address, Point of Contact for the invoice (Name, title, telephone number, e-mail and mailing address of point of contact).
2. Remit-to address (Name and complete mailing address to send payment).
3. Remittance name must match exactly with name on original order/contract. If the Remittance name differs from the Legal Business Name, then both names must appear on the invoice.
4. Invoice date.
5. Unique invoice #s for all invoices per vendor regardless of site.
6. NBS document number formats must be included for awards created in the NBS: Contract Number; Purchase Order Number; Task or Delivery Order Number and Source Award Number (e.g., Indefinite Delivery Contract number; General Services Administration number); or, BPA Call Number and BPA Parent Award Number.
7. Unique Entity Identifier (UEI) which is in the System for Award Management (SAM) and replaces the Dun & Bradstreet Data Universal Numbering System (DUNS) number.
8. Federal Taxpayer Identification Number (TIN). In those rare cases where a contractor does not have a UEI number or TIN, a Vendor Identification Number (VIN) must be referenced on the invoice. The VIN is the number that appears after the contractor's name on the face page of the award document.
9. Identify that payment is to be made using a three-way match.
10. Description of supplies/services **that match** the description on the award, by line billed.*
11. Freight or delivery charge must be billed as shown on the award. If it is included in the item price do not bill it separately. If identified in the award as a separate line item, it must be billed separately.
12. Quantity, Unit of Measure, Unit Price, Extended Price of supplies delivered, or services performed, as applicable, and that **match** the line items specified in the award.*

NOTE: If your invoice must differ from the line items on the award, please contact the Contracting Officer before submitting the invoice. A modification to the order or contract may be needed before the invoice can be submitted and paid.

B. The Contractor shall submit invoices to the Department of Treasury's Invoice Processing Platform (IPP) at <https://www.ipp.gov> with a copy to the approving official, as directed below.

The Contractor shall submit a copy of the electronic invoice to the following Approving Official (Contracting Officer) and Contracting Officer's Representative:

Approving Official: Contracting Officer

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Name- TBD at Award

Email Address- TBD at Award

Contracting Officer's Representative

Name- TBD at Award

Email Address- TBD at Award

II. Invoice Payment

A. Except as indicated in paragraph B., below, the due date for making invoice payments by the designated payment office shall be the later of the following two events:

1. The 30th day after the designated billing office has received a proper invoice.
2. The 30th day after Government acceptance of supplies delivered or services performed.

B. The due date for making invoice payments for meat and meat food products, perishable agricultural commodities, dairy products, and edible fats or oils, shall be in accordance with the Prompt Payment Act, as amended.

III. Interest Penalties

1. An interest penalty shall be paid automatically if payment is not made by the due date and the conditions listed below are met, if applicable.
2. A proper invoice was received by the designated billing office.
3. A receiving report or other Government documentation authorizing payment was processed and there was no disagreement over quantity, quality, or contractor compliance with a term or condition.
4. In the case of a final invoice for any balance of funds due the contractor for supplies delivered or services performed, the amount was not subject to further settlement actions between the Government and the Contractor.

B. Determination of interest and penalties due will be made in accordance with the provisions of the Prompt Payment Act, as amended, the Contract Disputes Act, and regulations issued by the Office of Management and Budget.

IV. PROVIDING ACCELERATED PAYMENT TO SMALL BUSINESS SUBCONTRACTORS, FAR 52.232-40 (November 2021)

(a) Upon receipt of accelerated payments from the Government, the Contractor shall make accelerated payments to its small business subcontractors under this contract, to the maximum extent practicable and prior to when such payment is otherwise required under the applicable contract or subcontract, after receipt of a proper invoice and all other required documentation from the small business subcontractor.

(b) The acceleration of payments under this clause does not provide any new rights under the

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Prompt Payment Act.

(c) Include the substance of this clause, including this paragraph (c), in all subcontracts with small business concerns, including subcontracts with small business concerns for the acquisition of commercial products or commercial services.

(End of Clause)

V. HHSAR 352.232-71 Electronic Submission of Payment Requests (February 2, 2022)

(a) Definitions. As used in this clause—

Payment request means a bill, voucher, invoice, or request for contract financing payment with associated supporting documentation. The payment request must comply with the requirements identified in FAR 32.905(b), “Content of Invoices” and the applicable Payment clause included in this contract.

(b) Except as provided in paragraph (c) of this clause, the Contractor shall submit payment requests electronically using the Department of Treasury Invoice Processing Platform (IPP) or successor system. Information regarding IPP, including IPP Customer Support contact information, is available at www.ipp.gov or any successor site.

(c) The Contractor may submit payment requests using other than IPP only when the Contracting Officer authorizes alternate procedures in writing in accordance with HHS procedures.

(d) If alternate payment procedures are authorized, the Contractor shall include a copy of the Contracting Officer’s written authorization with each payment request.

(End of Clause)